Qiskit Fall Fest 2026

Challenge 2: Make the Quantum Coin Biased — Complete Solution

The task requires one qubit, measurement, no more than four gates before measurement, and 100 shots. The circuits below use one RY rotation before measurement.

1. Circuit A — Bias toward 0

Gate: RY(π/3). Since RY(θ)|0■ = cos(θ/2)|0■ + sin(θ/2)|1■, P(0)=cos²(θ/2) and P(1)=sin²(θ/2). For θ=π/3, P(0)=0.75 and P(1)=0.25.

```
qc0 = QuantumCircuit(1, 1)
qc0.ry(np.pi/3, 0)
qc0.measure(0, 0)
result = simulator.run(transpile(qc0, simulator), shots=100).result()
counts0 = result.get_counts()
print(counts0)
plot_histogram(counts0); plt.show()
```

2. Circuit B — Bias toward 1

Gate: RY(2π/3). This gives P(0)=0.25 and P(1)=0.75, so outcome 1 should occur more often.

```
qc1 = QuantumCircuit(1, 1)
qc1.ry(2*np.pi/3, 0)
qc1.measure(0, 0)
result = simulator.run(transpile(qc1, simulator), shots=100).result()
counts1 = result.get_counts()
print(counts1)
plot_histogram(counts1); plt.show()
```

3. Circuit C — Stronger bias toward 0

Gate: RY(π/6). This gives P(0)≈0.933 and P(1)≈0.067.

```
qc2 = QuantumCircuit(1, 1)
qc2.ry(np.pi/6, 0)
qc2.measure(0, 0)
result = simulator.run(transpile(qc2, simulator), shots=100).result()
counts2 = result.get_counts()
print(counts2)
plot_histogram(counts2); plt.show()
```

4. Predicted results

Circuit	Gate	P(0)	P(1)	Approx. counts / 100
A	RY(π/3)	75%	25%	75 zeros, 25 ones
B	RY(2π/3)	25%	75%	25 zeros, 75 ones
C	RY(π/6)	93.3%	6.7%	93 zeros, 7 ones

The counts are approximate expectations; actual 100-shot counts fluctuate randomly around these probabilities.

5. Challenge question — answer

Quantum gates change the state amplitudes and relative phases of a qubit. On measurement, the probability of each outcome is the squared magnitude of its amplitude. For an RY(θ) gate applied to |0■, the state becomes cos(θ/2)|0■ + sin(θ/2)|1■, giving P(0)=cos²(θ/2) and P(1)=sin²(θ/2). Changing θ changes the probabilities and biases the coin toward 0 or 1. A phase-only gate such as T does not change immediate computational-basis probabilities by itself, though it may affect later interference.

Setup: If needed, run `%pip install qiskit qiskit-aer matplotlib numpy` in a notebook cell. Qiskit was not installed in the preparation environment, so actual simulator counts are generated when you run the notebook.